

Quick-thinking, Confident, Communicative, and Collaborative: Fundamentals of Applied Improvisation for (Bio)statisticians and Data Scientists

Presented by Richard C. Zink, Ph.D

Principal Research Fellow

JMP Statistical Discovery LLC

Adjunct Professor, Biostatistics

Adjunct Assistant Professor, Public Health Leadership and Practice

The University of North Carolina at Chapel Hill

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ABOUT THE INSTRUCTOR:



Richard C. Zink is a biostatistician and software developer who has spent more than 20 years in and around clinical trials and medical product development. He is author, editor, and contributor to 10 books on statistical topics in clinical trials and clinical research, and he is currently working on a data visualization book featuring life sciences applications. He was awarded the distinction of Fellow of the American Statistical Association in 2020. Richard holds a Ph.D. in Biostatistics from the Gillings School of Global Public Health at the University of North Carolina at Chapel Hill, where he serves as Adjunct Professor of Biostatistics, providing seminars related to his research, teaching classes on clinical trials and data visualization, and collaborating in student research. He also serves as Adjunct Assistant Professor for the Department of Public Health Leadership and Practice, where he offers courses in applied improvisation to public health professionals. Richard teaches courses and

workshops on applied improvisation locally and at conferences for scientific and professional organizations, mostly notably for statisticians and data scientists.

ABSTRACT:

(Bio)statisticians and data scientists are technically proficient. This should come as no surprise, since their education focuses on developing quantitative scientists of the highest order. However, this leaves little opportunity for the development of interpersonal skills which are often just as important to one's success, particularly in a multidisciplinary environment. Ironically, many of the skills leading to technical success, such as planning, meticulousness, and specificity can stifle innovation, limit awareness, and inhibit communication. But all is not lost! Applied improvisation is a novel (and fun) way to pursue and practice interpersonal skills that are un- or under-developed.

Applied improvisation, the application of the principles and tools of improvisational theater in non-theatrical settings, has been used for decades in business schools to assist students in developing and expanding quick thinking, confidence, communication, and collaboration. Similar training is used to help

clinicians, healthcare workers, and scientists better engage their patients, collaborators, and non-technical colleagues. Improvisation skills aid leaders to be eager servants to their team, exhibit vulnerability, communicate more effectively, and create an atmosphere where individuals can think and share freely.

This half-day professional development course introduces individuals to applied improvisation from the perspective of a fellow quantitative scientist. Through several physical and verbal exercises, participants explore interpersonal skills such as creativity, spontaneity, adaptability, courage, and storytelling. Briefings examine how exercises reinforce various competencies and how they relate to everyday workplace pursuits. Resources for further reading and training are provided.

There is no requirement to be funny, though funny often naturally happens! No experience is required, just an open mind, an eagerness to participate, and a willingness to take risks in a supportive environment. If you want to increase confidence, react more effectively to unexpected events, engage your colleagues with active listening, and build ideas through active collaboration, then this short course is for you!

Learning Objectives

1. Recognize how applied improvisation develops and reinforces interpersonal skills appropriate for the workplace.
2. Summarize the skills various physical and verbal exercises emphasize and how exercises can improve communication, collaboration, and spontaneity.
3. Implement feedback and peer evaluation to improve listening, support others, and communicate more effectively.
4. Critique the effectiveness of individual physical and verbal exercises for personal skill development.

Course Description

Group and paired interactive physical and verbal exercises explore the topics listed above. Exercises are first demonstrated by the instructor and enthusiastic volunteers. Students engage in exercises where they receive feedback and coaching to more effectively engage one another, react spontaneously, and fully commit to the tasks at hand. Briefings and group discussion examine how the exercises reinforce various practical competencies and how they relate to everyday workplace pursuits. In many cases, exercises are modified to add additional challenges or task-loading requiring individuals to stretch their comfort zones, recall details, commit to decisions, and support their teammates even in cases when missteps are made. Students have the opportunity to evaluate their peers when not participating in exercises. For exercises addressing particular topics or competencies, group discussion critiques the pros and cons of the various games to understand the features that are most effective for professional skill development. To end the course, resources for further reading and training are provided and discussed. Students are encouraged to share insights they gained from the course and how these skills can be incorporated into their daily lives.

The importance of improvisation training for statisticians is discussed in "Improvise to improve: Why statisticians need to live in the moment", available in the September Issue of [Significance](#) magazine.

Location: Trinity Cathedral Conference Center, 2300 Euclid Ave. Cleveland, OH 44115

Time: The course will run from 8:30 am to 12:30 pm and then lunch. Registration will start at 8:00. Lunch is included in the registration cost.

Cost:

Full time student	\$10
ASA/ASQ	\$35
Non-member	\$60

Registration and Payment: <https://forms.gle/XcefsUn1r6R98Zj16>